

20th March 2023, 11:30 am – 12:30 pm

Course Code: CS2005	Course Name: Database Systems
Instructor Name: Anam Qureshi	
Student Roll No:	Section:

Instructions:

- Return the question paper. Don't write anything on the question paper, except your Roll # & Section #.
- Read each question completely before answering it. There **are 4 questions and 2 pages**.
- In case of any ambiguity, you may make assumptions. But your assumptions should not contradict any statement in the question paper.
- All the answers must be solved according to the sequence given in the question paper.
- This paper is subjective. Write the answers only on the answer sheet.

Time: 60 minutes.

Max Marks:15

Question 1:

[4 marks]

Consider the following initial valid state of an HR database. Identify the relational integrity violations if the following updates are made. Also provide the solution in case of any violation.

Note: The domain of dept_id is {1,2,3,4,5}.

- Insert into COUNTRIES values('1', 'Newzealand');
- Insert into EMPLOYEES values('EMP235', 'Ali', 'Agha', 'aliahmed@gmail.com', '18/9/2013', 6);
- Delete the DEPARTMENT's tuple with Dept_id=4;
- Delete the JOB_HISTORY tuple with Emp_id='EMP456';

EMPLOYEES

Emp_id	Fname	Lname	Email	Hire_date	Salary	Dept_id
EMP123	Ahmed	Zafar	Ahmed.zafar@gmail.com	12-24-2018	56000	4
EMP234	Sophia	Ahmed	Sohaib.ahmed@gmail.com	09-23-2017	76000	5

DEPARTMENTS

Dept_id	Dept_name	Loc_id
4	Electronics	45
5	Civil	67

LOCATIONS

Loc_id	Postal_code	City	State_province	Country_id
45	78100	Khi	Sindh	1
67	73100	Khi	Sindh	1

COUNTRIES

Country_id	Country_name
1	Pakistan
2	Turkey

JOB_HISTORY

Emp_id	Start_date	End_date	Dept_id
EMP123	12-24-2018	InService	4
EMP234	09-23-2017	12-23-2018	5

Question 2:

[3 marks]

The Dawlance Company maintains its record of Products sale in File Based System at Regional Offices. Now they want to develop a Mobile App to monitor production and to compete the demand and supply issues in the market for their products. Briefly explain how the DBMS will prove beneficial with respect to their business demands, also discuss the issue(s) in their current record-keeping system.

Question 3:

[3 marks]

Ahmed has decided to start his own business of web hosting. He has planned to provide the web hosting service globally. Suggest Ahmed about the choice of client-server architecture; also justify your answer with some strong arguments.

Question 4:**[5 marks]**

You are designing a database for KW Humane Society. The result is the following set of relations where the type of each relations attribute is given following the attribute (e.g., ID: integer):

Animals (ID: integer, Name: string, PrevOwner:string, DateAdmitted:date, Type: string)

Adopter (SIN: integer, Name: string, Address: string, OtherAnimals: integer)

Adoption (AnimalID: integer, SIN: integer, AdoptDate: date, ChipNo: integer)

Where

- The primary keys are underlined.
- Animals store information about the animals currently at the Humane Society. Each is given an ID, and their name together with the SIN of their previous owners (attribute PrevOwner), and their date of admission is recorded. Type refers to the type of animal (dog, cat, etc)
- Adopter is the relation that holds information about animals' adopters. The attributes are self-descriptive, except OtherAnimals which records the number of other animals that the adopter currently has at home.
- AnimalID in Adoption refers to ID of Animals. Similarly, SIN in Adoption refers to the SIN of Adopter. Attribute ChipNo stores the number on the microchip that is implanted on the animal for tracking. Owner in Animals refers to the SIN of Adopter(in this case the previous adopter).

Formulate the following queries in SQL:

- 1) Retrieve the total number of dogs that were brought to the Humane Society on 18 April 2019.
- 2) List the name of the adopter who has adopted every type of animal.
- 3) For each animal type, list the animal type and total number of adoptions on 14 June 2019.
- 4) List the types of animals who have not had any adoptions.
- 5) For each adopter who has made at least two adoptions, list their names and addresses.

Good Luck